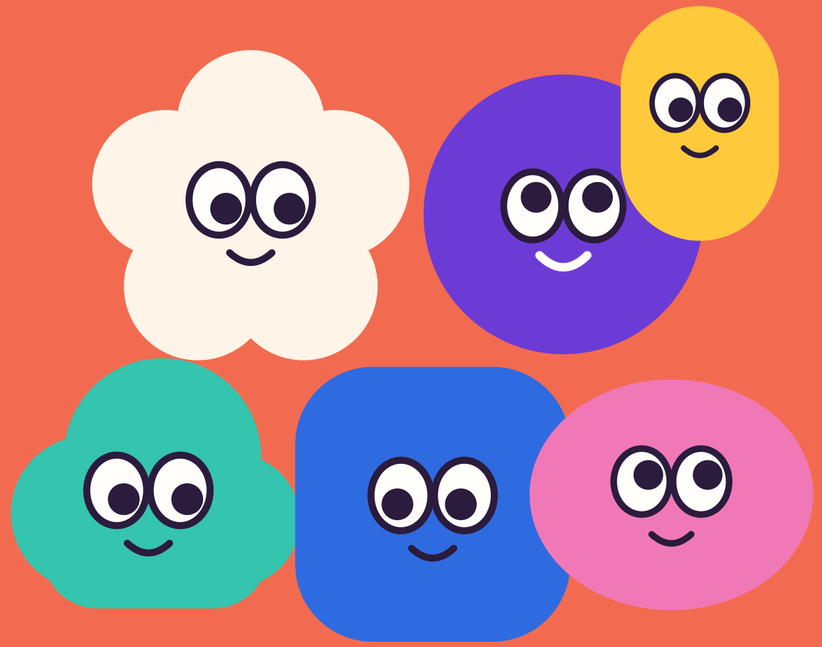
JEF YOUTH FILM FESTIVAL · XL-MEDIALAB 2027

Bachelor's Thesis

From research to an interactive installation for children aged 6–12.

Ibe · Senne · Tjorven — Devine, Howest



THE BRIEF

JEF wants one installation that works for every child, all day, without staff

6–12
years old: the age range of the visitors

±5,000
visitors during the festival

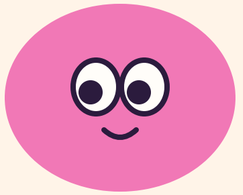
18
days: 28 January – 14 February 2027

0
technical staff on site to explain or restart it

Our challenge: how might we create an immersive multiplayer experience that a 6-year-old can understand, while still being challenging for a 12-year-old?

OUR AUDIENCE

Children aged 6–12, visiting in groups, staying about a minute



6-year-olds often cannot read yet	→	**Everything is shown, nothing is written**
12-year-olds want a real challenge	→	**The difficulty has to grow with the player**
They differ a lot in height and motor skills	→	**Being tall or small must not decide who wins**
They arrive with classes and families	→	**Many children must be able to play at once**
They come and go freely	→	**No start button, no queue, no staff**

OUR RESEARCH



Before designing anything, we wanted to understand how children play

Where we looked

Play research
Children's play, motivation and development

Visitor studies
How people behave at museum installations

Precedents
Existing interactive installations and games

The client
The briefing from JEF

INSIGHT 1 OF 5

Children decide in seconds whether to stay



WHAT WE FOUND

±60 s

Children spend about a minute at one installation on average, and the first seconds decide whether they stay.

WHAT IT MEANS FOR US

A reaction within 10 seconds

No long explanation or tutorial. Everything has to be self-explanatory.

Sources: Dancstep, T., & Gutwill, J. P. (2019), 906 children aged 8–13 · Hornecker, E., & Stifter, M. (2006). *OzCHI '06*, 135–142.

INSIGHT 2 OF 5

Their body is the best controller



WHAT WE FOUND

Body first

Children explore cause and effect directly with their bodies. Showing them how it works makes them explore less. Hand-held controllers fail across ages, because motor skills differ.

WHAT IT MEANS FOR US

The body is the only controller

No buttons, no objects to hold, no text to read.

Sources: Price, Sakr & Jewitt (2016). *Interacting with Computers, 28*(5) · Bonawitz et al. (2011). *Cognition, 120*(3) · Hornecker & Stifter (2006). *OzCHI '06*.

INSIGHT 3 OF 5

Groups shape the experience



WHAT WE FOUND

Honeypot

Groups come and go at once, because children follow each other. But people who watch others play are drawn in to join: the honeypot effect.

WHAT IT MEANS FOR US

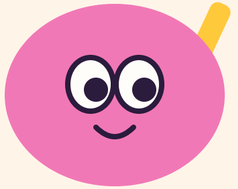
Many players, visible from afar

Many children play at once, the play is visible from across the room, and cooperation is rewarded, never required.

Source: Brignull, H., & Rogers, Y. (2003). Enticing people to interact with large public displays in public spaces. *Proceedings of INTERACT 2003*.

INSIGHT 4 OF 5

Children care more about what they made



WHAT WE FOUND

Made by me

Children choose to draw more often when their work is kept. The best-rated installations show what visitors made.

WHAT IT MEANS FOR US

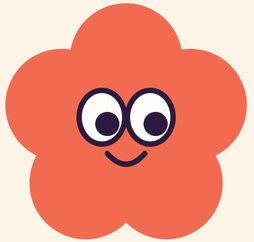
Make it yours

Children draw their own character on paper or an iPad. It joins the world and stays visible after they leave.

Sources: Liu, Y., & Yang, F. (2026). *Psychological Science, 37*(8) · Hornecker, E., & Stifter, M. (2006). *OzCHI '06*, 135–142.

INSIGHT 5 OF 5

Children remember the high point



WHAT WE FOUND

The peak

Children stay motivated by short, clear goals and instant feedback. And people remember an experience mostly by its peak and its end.

WHAT IT MEANS FOR US

A moment to be proud of

Short goals that take less than a minute, and a clear high point children will tell their parents about.

Sources: Malone, T. W. (1981). *Cognitive Science, 5*(4), 333–369 · Kahneman, D., et al. (1993). *Psychological Science, 4*(6), 401–405.

FROM INSIGHTS TO PRINCIPLES

Six design principles

1 Instant feedback

We instantly see action → reaction.

2 No controller needed

No buttons or levers, just movement.

3 Explain without words

Learn by trying, understand it instantly.

4 A moment worth remembering

A clear peak, or a goal reached.

5 Cooperation is rewarded, not required

Teaming up helps, playing alone still works.

6 Everyone sees your creation

Your drawing appears on the screen.

Insight 1 → 1 · Insight 2 → 2 and 3 · Insight 3 → 5 · Insight 4 → 6 · Insight 5 → 4

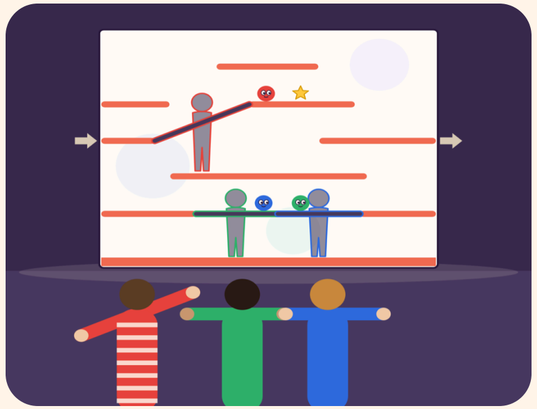
OUR CONCEPT (FOR NOW) · [WORKING TITLE]

Your shadow becomes the bridge



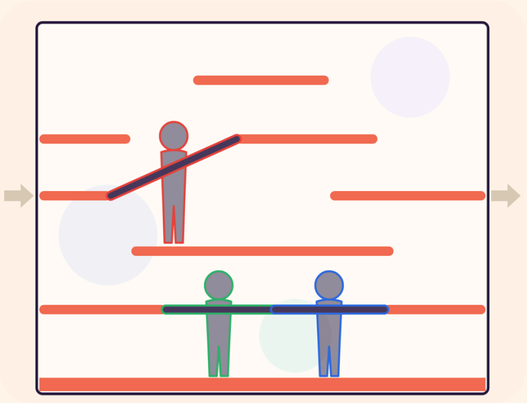
1 Draw a character

On paper or an iPad. It walks into the world on the wall.



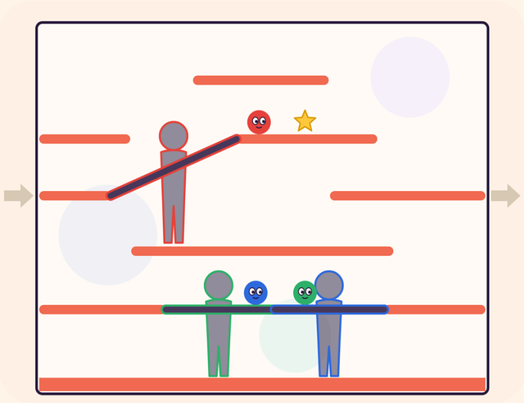
2 Step in

A camera sees you, and your shadow appears in the world, linked to your drawing.



3 Build with your arms

Arms out make a bridge, a tilted arm a ramp. Wide gaps need two children.



4 Reach the stars

Characters cross, climb and collect stars. Collect three for a special effect.

No game over: a character that misses tumbles one level down. Theme and look are still open.

▶ Inspired by this reel

NEXT: TESTING AND BUILDING

What we still need to find out



One continuous world, or levels?

How: Children play both; we measure time, replays and smileys

Do children understand the gestures without explanation?

How: Observe the first 10 seconds

How many children can play at one wall?

How: Play sessions with small groups

What materials and hardware do we need?

How: Build one wall as a prototype: depth camera, projector and Unity

WHAT'S NEXT

Our to-do list

- [x] **Divide the team roles**
  Based on our strengths: UX, tech and design.
- [ ] **Make a full planning**
  From now up to the festival.
- [ ] **Set deadlines**
  Clear milestones we can hold each other to.
- [ ] **Develop the concept further**
  Theme, look, levels and the peak moment.
- [ ] **Figure out how to test and build it**
  Test setup with children, hardware and a first prototype.

THANK YOU

Spread your arms. The world is waiting.

Questions?

